BTS (C) – VIII – 10 – 001 – Q

B.Tech Degree VIII Semester Examination, April 2010

ME 801 OPERATIONS MANAGEMENT
(2006 Scheme)

Time: 3 Hours | Maximum Marks: 100

PART – A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Explain algorithm for finding critical path.
(b) Compare CPM and PERT.
(c) What is "Product Life Cycle".
(d) Explain "Bin Card System".
(e) Describe "Gantt Chart".
(f) What are the functions of Despatch.
(g) What are the principles of material handing.
(h) What is "Depreciation"

PART – B

(All questions carry ***EQUAL*** marks)

(15 x 4 = 60)

II. Draw the PERT network and calculate project completion time. The following informations are available.

Predecessor activities	A	B	C	D	E	F	G	H	I	J	K	L
Successor activities	D F	L	B E	G H	G H	I K	I K	L	L	G H	-	-
Duration (week)	8	9	10	7	6	9	4	3	3	12	10	16

OR

III The precedence relationship with Node numbers of a maintenance job are given as follows: (i) Draw the arrow diagram and show the critical path (ii) indicate the slack times for jobs (3, 5) and (7, 8) (ie.) for 'd' and 'l'.

Job	Node Number	Estimated Duration (Days)
a	(1, 2)	2
b	(2, 3)	3
c	(2, 4)	4
d	(3, 5)	5
e	(3, 6)	10
f	(4, 6)	6
g	(4, 7)	3
h	(5, 8)	8
k	(6, 8)	7
l	(7, 8)	3

(Turn Over)

IV. Write notes on: (i) Interchangeability (ii) Standardisation (iii) Product design

OR

V. A concern is engaged in casting of carburetors. It's demand from the automobile industries per year is 50,000. The cost of set up is Rs.1000/-. There are ten workers engaged, each on wage rate of Rs. 15/- per day. The overhead cost is Rs.100/- per day. The daily production capacity is 200/-. The material cost for each carburetor is Rs.10/-. The annual rate of depreciation, insurance, taxes and storage cost, etc. is 20% of unit cost. The supply should be instantaneous and no shortages are permitted.

Find: (i) The economic lot size (ii) The number of runs (iii) Duration of each run

VI. Determine an optimum sequence to process the various types of fan blades each day from the following information so as to minimize the total elapsed time.

Type of fan Blade	Number to be Processed in each day	Processing time on Machine A (Minutes)	Processing time on Machine B (Minutes)
1	4	4	8
2	6	12	6
3	5	14	16
4	2	20	22
5	4	8	10
6	3	18	2

Also workout the total elapsed time for an optimum square. What is the total idle time on Machine A and on Machine B.

OR

VII. We have five jobs each of which must go through three machines A, B and C in the order A, B, C. Processing time in hours are given. Determine a sequence for the five jobs that will minimize the total elapsed time. Find also the idle time for machine A, B and C.

Job i	Processing Time Ai	Bi	Ci
1	16	10	8
2	20	12	18
3	12	4	16
4	14	6	12
5	22	8	10

VIII. (a) Explain the factors to be considered for plant location.

(b) What are the different types of plant layouts. Explain any two in detail.

OR

IX. (a) Differentiate between preventive maintenance and breakdown maintenance.

(b) How can we find the optimum period for replacement? Show graphical representation.

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